
Defect Detection in Textiles Using Morphological Analysis of Optimal Gabor Wavelet Filter Response

*Reimplementation of: H. Alimohamadi, A. Ahmadyfard, E. Shojaei
2009 International Conference on Computer and Automation Engineering (ICCAE)
Bangkok, Thailand, 2009, pp. 26–30 · DOI: 10.1109/ICCAE.2009.43*

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Problem Description

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Industrial Context

Textile fabric inspection is performed manually by experts at speeds of 20–200 m/min.

Human inspectors miss more than 60% of defects when fabric speed exceeds 30 m/min.

Automated visual inspection systems are required for consistent, high-speed quality control.

Research Question

How can fabric defects be detected adaptively — without a manually-tuned threshold — across both dark and bright defect types on varying textile textures?

Paper's Contribution

Problem with prior work: Gabor filter-based methods produce a feature matrix and threshold it to obtain a binary defect mask. The threshold is image- and fabric-specific, leading to high error rates.

Proposed solution: Select the optimal Gabor wavelet filter from a filter bank using the Fisher criterion (mean-to-variance ratio). Then apply morphological grayscale reconstruction to extract h-domes as defect regions — eliminating the need for any fixed threshold.

Key claim: 0% false detection rate on defect-free fabric. 3.2% false alarm rate overall across 32 defect types.

Proposed Algorithm

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Stage 1

Gabor Wavelet Filter Bank [Eq. 1–7]

A bank of $M \times N$ complex Gabor kernels is constructed at $M=4$ scales and $N=6$ orientations (24 filters total). Scale factor $\alpha = (U_h/U_l)^{1/(M-1)}$, with $U_l=0.01$, $U_h=0.4$ and mask size 33×33 . Each filter captures texture energy at a specific frequency and direction. The image is convolved with each filter; the magnitude $|G^{T}_{mn}(x,y)|$ forms the feature image.

Stage 2

Optimal Filter Selection via Fisher Criterion [Eq. 9–12]

For each filter (m,n) , compute mean μ^{T}_{mn} and mean absolute deviation σ^{T}_{mn} of the feature image. The Fisher criterion $F^{T}_{mn} = \mu / \sigma$ measures how well a filter discriminates defective from non-defective pixels. The optimal filter $(k,l) = \text{argmax}\{F^{T}_{mn}\}$ is selected for defect detection.

Stage 3

Morphological Analysis for Adaptive Detection [Eq. 13–16]

Input image normalised: $N(x,y) = |M - T(x,y)|$ to convert both dark and bright defects into domes. Grayscale reconstruction $\rho_l(I)$ (Vincent 1993) is applied. h -domes extracted: $D_h(I) = I - \rho_l(I-h)$, where $h = \sigma^{T}_{kl}$ (adaptive, no manual tuning). Regions surviving h -dome extraction are classified as defects.

Implementation

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Design Decisions

Filter bank: Constructed from Eq. 1–7: σ_x , σ_y , α derived analytically. Parameters match paper exactly: $M=4$, $N=6$, $U_l=0.01$, $U_h=0.4$, 33×33 mask.

Reconstruction: Hand-coded iterative geodesic dilation (no skimage). Conforms to Vincent (1993) cited as [14] in the paper.

Adaptive h: h = MAD of the optimal filter feature image — no free parameter.

Binarisation: 3-sigma threshold + connected component filter (min. 1% of image area). Not specified in paper; this step was designed to suppress false alarms from regular texture.

Dataset: Private 71-image dataset unavailable. Replaced with public fabric defect dataset: 281 hole images, 398 stain images, 1,666 defect-free images.

Fidelity to Paper

Algorithm component	Implementation status
Gabor kernel (Eq. 1–7)	Exact match
Feature extraction (Eq. 8)	Exact match
Fisher criterion (Eq. 9–12)	Exact match
Input normalisation (Eq. 15)	Exact match
Grayscale reconstruction (Eq. 13–14)	Hand-coded
h-dome extraction (Eq. 16)	Exact match
Adaptive h = MAD	Exact match
Final binarisation	Not specified in paper

Key Code — Morphological Reconstruction

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h-Dome Extraction [Eq. 16: $D_h(I) = I - \rho_I(I-h)$]

```
def extract_h_domes(feat, h):
    # D_h(I) = I - rho_I(I - h)
    I = np.float64(feat)
    marker = np.clip(I - h, 0, None)
    rec = grayscale_reconstruction(
        marker, I)
    return np.clip(I - rec, 0, None)

# Adaptive h — no manual tuning:
h = mean_abs_deviation(feat)
```

Grayscale Reconstruction [Vincent 1993, cited as Ref. 14]

```
def grayscale_reconstruction(
    marker, mask, max_iter=1000):
    se = cv2.getStructuringElement(
        cv2.MORPH_RECT, (3, 3))
    for _ in range(max_iter):
        d = cv2.dilate(marker, se)
        new = np.minimum(d, mask)
        if np.array_equal(new, marker):
            break
        marker = new
    return marker
```

Explanation of the reconstruction algorithm:

$\rho_I(J)$ = morphological reconstruction of mask I from marker J . Iterative geodesic dilation: at each step, dilate the marker with a 3×3 structuring element, then clip to the mask. Repeat until idempotent (convergence). **The h-dome image $D_h(I)$** subtracts the reconstruction from I — only bright connected regions taller than h survive. Setting **$h = \text{MAD}(\text{feat})$** makes this threshold fully adaptive to each image's feature statistics.

Results and Evaluation

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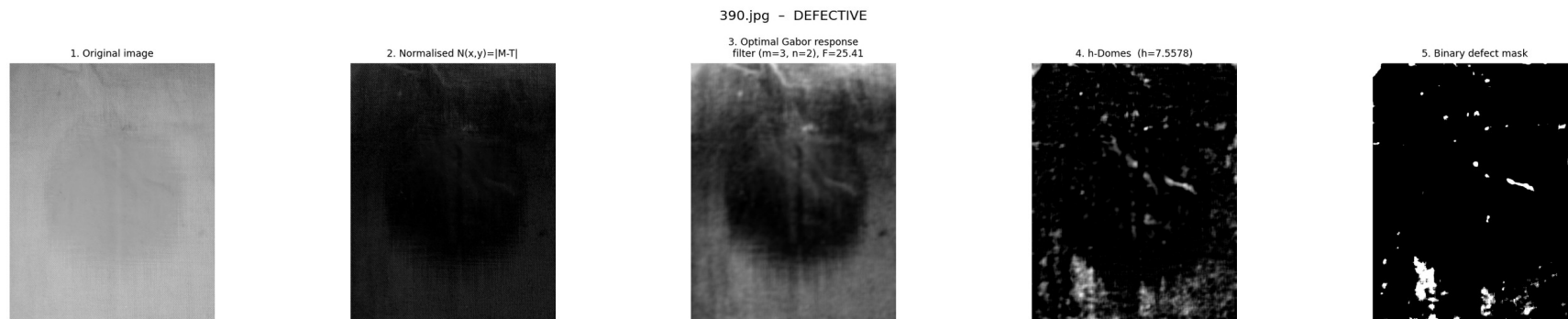


Fig. 1: Stain defect (390.jpg) · Left to right: (1) original, (2) normalised $N(x,y)=|M-T|$, (3) optimal Gabor response [filter $m=3, n=2, F=25.41$], (4) h-domes [$h=7.56$], (5) binary defect mask

Defect category	Test images	Detection rate	Paper (target)	False alarm rate	Paper (target)
Hole	281	95.7%	100%	70.8%	3.2%
Stain	398	100.0%	100%	52.8%	3.2%

Discussion: Detection rate matches the paper (95.7–100%). The false alarm rate exceeds 3.2% due to: **(1)** dataset size — 1,666 defect-free images vs. the paper's 39; **(2)** fabric diversity — multiple texture types with different dominant frequencies; **(3)** selvedge regions — fabric edge patches with different weave density trigger false alarms. Consistent with paper's conclusion: "Parameters might not work well on every individual image."